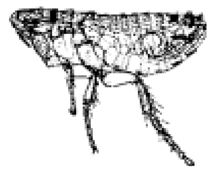
flea



NEWS 77 *December 2015*

http://esanetworks.org/

FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book and Literature Reviews, Biography, Hypotheses, and Anecdotes. Send to:

R. L. Bossard, Ph.D.
Editor, Flea News
bossardtech@gmail.com

Organizers of the Flea News Network are Drs. R. L. Bossard and N. C. Hinkle.

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Assistant Editor J. R. Kucera, M.S.

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Editorial

Dear *Flea News* Reader,

Thank you to everyone who answered the *Flea News* survey, sent out after the previous *Flea News*. Every response was carefully reviewed. All the features of current *Flea News* were enjoyed by some readers, although certain features, especially the list of flea citations, were helpful to everyone.

One idea is to separate the citations into topics, such as flea taxonomy, ecology, evolution, control, and disease. Though it is currently not practical to separate the hundreds of citations that appear in a volume of *Flea News*, the citations in the Featured Research are separated by topic in this volume, emphasizing new species descriptions (see the Announcements in this *Flea News* for the tabulating of flea species diversity).

A second idea is to create a separate website for *Flea News*. For now, *Flea News* will stay with the Networks website, generously hosted by the Entomological Society of America, as well as emailing *Flea News* upon request. The editors thank the Entomological Society of America for their support.

The editors were pleased to find that readers of *Flea News* found the format satisfactory, and that *Flea News* continues to fulfill its purpose as a touchstone of communication among scientists interested in fleas, as well as contribute to medical-veterinary entomology, disease control, and the study and conservation of biodiversity. Please send to the editors additional comments.

Thank you for your support of *Flea News*.

Yours in fleas,
R.L. Bossard, N.C. Hinkle
Editors, *Flea News*

***

Announcements

Dr. Lane Foil of LSU is looking for a *Ctenocephalides felis* colony, after the demise of his former lab colony. If you can assist him with re-establishing a colony, please contact him at:

Lane Foil
Pennington Regents Chair for Wildlife Research
Subject Editor, J. Med. Entomol.
110 Union Sq.
404 Life Sciences Building-LSU
Baton Rouge, LA 70803
225-578-1825
Fax: 225-578-7307
LFoil@agcenter.lsu.edu

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Total Number, and an Updated Listing, of Flea Species of the World

Ongoing questions are: how many flea species are there in the world, and what are they? Could we compile an updated list of flea species of the world?

The book "World Species Index: Siphonaptera: A Comprehensive Catalog of Fleas of the World", that is a listing of flea species (reviewed in the previous *Flea News* 76, June 2015), is mostly a copy of Sergei Medvedev's excellent online database (http://www.zin.ru/Animalia/Siphonaptera/taxfind2.asp), but without citation of the database, wrote Sergei. That website lists 2,333 flea species, but the database has not been updated for several years. The most recent citation I found there was Lewis and Stone (2001).

The following lists of species are from Michael Hastriter (with additions from the editors):

Taxa amended or described since 1 November 2009*

2010
Trochilopsylla n. gen. Beaucournu and Gonzalez-Acuna, 2010
Trochilopsylla torresmurai Beaucournu and Gonzalez-Acuna, 2010

2011
Agastopsylla guzmani Beaucournu, Moreno and Gonzalez-Acuna, 2011
Delostichus degus Beaucournu, Moreno and Gonzalez-Acuna, 2011
Ectinorus (Ectinorus) spiculatus Hastriter and Sage 2011
Ectinorus (Onychius) onychius New Combination (Hastriter and Sage, 2011)
Ectinorus (Onychius) onychius angularis = Ectinorus angularis New Combination (Hastriter and Sage, 2011)

Ectinorus (Onychius) onychius deplexus = *Ectinorus deplexus* New Combination (Hastriter and Sage, 2011)
Ectinorus (Onychius) onychius fueginus = *Ectinorus fueginus* New Combination (Hastriter and Sage, 2011)
Gryphopsylla (Migrastivalius) jacobsoni segregata Beaucournu and Sountsov, 1999 elevated to full species *Gryphopsylla (Migrastivalius) segregata* Beaucournu and Sountsov, 1999 (Beaucournu and Wells, 2011)
Smitictenophthalmus n. subgen. Beaucournu, 2011

2012
Dasypsyllus (Neornipsyllus) patagonicus Beaucournu, Munoz-Leal and Gonzalez-Acuna, 2012
Eospilopsyllus n. gen. Beaucournu and Perrichot (in Perrichot, Beaucournu and Velten, 2012)
Eospilopsyllus kobberti Beaucournu and Perrichot (in Perrichot, Beaucournu and Velten, 2012)
Lagaropsylla makay Laudisoit, Prie, and Beaucournu, 2012
Neotyphloceras crassispina chilensis Jordan, 1936 elevated to full species: *Neotyphloceras chilensis* Jordan, 1936 (Sanchez, et al., 2012)
Nestivalius sulawesiensis Mardon and Durden, 2003 [amendment?]
Nosopsyllus (Nosopsyllus) atsbi Beaucournu, Meheretu and Laudisoit, 2012 (in Beaucournu, Meheretu, Welegerima, Mergey and Laudiisoit, 2012)
Tunga bonneti Beaucournu and Gonzalez-Acuna, 2012 (in Beaucournu, Mergey, Munoz-Leal and Gonzalez-Acuna, 2012)
Wilsonipsylla n. gen. Hastriter, 2012
Wilsonipsylla spinicoxa Hastriter, 2012

2013
Ectinorus (Ectinorus) insignis Beaucournu and Gonzalez-Acuna, 2013 (in Beaucournu, Belaz, Munoz-Leal and Gonzalez-Acuna, 2013)
Lentistivalius philippinensis Hastriter and Bush, 2013
Pygiopsylla spinata Holland, 1969 is a junior synonym of *Pygiopsylla hoplia* Jordan and Rothschild, 1922 (Hastriter and Easton, 2013)
Striopsylla bifurcata Hastriter and Easton, 2013
Striopsylla clarki Hastriter and Easton, 2013
Striopsylla flanneryi Hastriter and Easton, 2013
Striopsylla bifurcata Hastriter and Easton, 2013
Striopsylla clarki Hastriter and Easton, 2013
Striopsylla flanneryi Hastriter and Easton, 2013

2014
Medwayella independencia Hastriter and Bush, 2014
Mirzapsylla n. subgen. Hastriter, 2014
Neotyphloceras crackensis Sanchez and Lareschi, 2014
Neotyphloceras pardinasi Sanchez and Lareschi, 2014
Orthopsylloides (Orthopsylloides) mardoni Hastriter, 2014 (female unknown)
Orthopsylloides (Orthopsylloides) maracaballa Hastriter, 2014
Orthopsylloides (Orthopsylloides) abacetus baiyerensis George and Beaucournu, 1995 is synonym of *Orthopsylloides (Orthopsylloides) abacetus* (Jordan and Rothschild, 1922) (Hastriter, 2014)
Orthopsylloides (Orthopsylloides) orthodactylus owiensis Holland, 1969 is synonym of *Orthopsylloides (Orthopsylloides) orthodactylus* Holland, 1969 (Hastriter, 2014)

Orthopsylloides (Orthopsylloides) penguinata Hastriter, 2014
Orthopsylloides (Orthopsylloides) uncinata Hastriter, 2014
Orthopsylloides (Mirzapsylla) andersoni George and Beaucournu, 1995 New combination (Hastriter, 2014)
Orthopsylloides (Mirzapsylla) rawlinsi Hastriter, 2014 (female unknown)
Orthopsylloides (Mirzapsylla) repanda Hastriter, 2014
Orthopsylloides (Mirzapsylla) truncata Hastriter, 2014
Orthopsylloides (Mirzapsylla) whitingi Hastriter, 2014
Parastivalius avicularius Hastriter, 2014
Parastivalius bidenticulatus Hastriter, 2014
Parastivalius chelaformis Hastriter, 2014
Parastivalius longipalpus Hastriter, 2014
Parastivalius novaebritainiae Hastriter, 2014
Parastivalius papillatus Hastriter, 2014
Parastivalius phocaceus Hastriter, 2014
Parastivalius spinatus Hastriter, 2014
Parastivalius tridenticulatus Hastriter, 2014
Tsaractenus clavator Beaucournu and Laudisoit, 2014

2015
Atopopsyllus cionus, n. gen., n. sp. Poinar, 2015
Ctenidiosomus austrinus Lopez-Berrizbeitia, Hastriter, and Diaz (in Lopez-Berrizbeitia, Hastriter, Barquez, and Diaz, 2015)
Plocopsylla linardii Sanchez, Beaucournu and Lareschi, 2015
Traubia (Traubia) durdeni Hastriter, 2015
Traubia (Traubia) pilgrim Hastriter, 2015
Traubia (Traubia) traubi Hastriter, 2015
Traubia (Traubia) wilhelmensis Hastriter, 2015
Traubia (Kuruopsylla) n. subgen. Hastriter, 2015
Traubia (Kuruopsylla) eaglini Hastriter, 2015
Traubia (Kuruopsylla) mulcahyi Hastriter, 2015
Traubia (Kuruopsylla) o'harai Hastriter, 2015
Traubia (Kuruopsylla) ungerechti Hastriter, 2015

2016 (in review)

Taxa described in 2009 and before that were omitted from Lewis (2009)*

Amphalius spirataenius mengdaensis Wu and Wang, 2002
Aphropsylla truncata Hastriter, 2009
These five Cediopsylla taxa were omitted from Lewis 2009 list:
    Cediopsylla inaequalis inaequalis (Baker, 1895)
    Cediopsylla inaequalis interrupta Jordan, 1925
    Cediopsylla simplex (Baker, 1895)
    Cediopsylla spillmanni Jordan, 1930
    Cediopsylla tepolita Barrera, 1967
Consobrinopsyllus n. subgen. Duchemin, 2003
Dasypsyllus (Neornipsyllus) huinayensis Beaucournu, Ardiles and Gonzalez-Acuna, 2009)
Dasypsyllus (Neornipsyllus) tapaculensus Beaucournu, Ardiles and Gonzalez-Acuna, 2009)

Dorcadia tiani Liu, Fu, and Wu, 2003
Ectinorus (Ectinorus) hirsutus Hastriter, 2009
Ectinorus (Ectinorus) lareschiae Hastriter and Sage, 2009
Ectinorus (Ectinorus) morenoi Hastriter and Sage 2009
Farhangia quattuordecimdentata Mardon and Durden, 2003
Farhangia sedecimdentata Mardon and Durden 2003
Gryphopsylla (Destivalius) mjoebergi (Jordan, 1926) (moved from *Lentistivalius* to *Gryphopsylla* by Mardon, 1981)
Hoogstraalia novaeguineae Holland, 1969
Jellisonia tiptoni (Mendez and Altman, 1960) (*Kohlsia tiptoni* Mendez and Altman, 1960 transferred to the genus *Jellisonia* (Hastriter, 2004)
Kohlsia zyanya Acosta, et al., 2009
Kohlsia ahuacatlan Acosta, et al., 2009
Lagaropsylla mytila Hurka, 1984 is junior synonym of *Lagaropsylla putila* Jordan and Rothschild, 1921 (Beaucournu and Kock, 1994)
Macrostylophora (Macrostylophora) perplexa Beaucournu and Sountsov, 1999
Macrostylophora (Songshupsylla) cuiae jiangkouensis Li and Huang, 1979 (elevated to full species *Macrostylophora (Songshupsylla) jiangkouensis* Li and Huang, 1979 (Hou and Liu, 2003)
Macrostylophora (Macrostylophora) theresae Durden and Beaucournu, 2006 should be placed within the subgenus *Songshupsylla*, not *Macrostylophora* [*Macrostylophora (Songshupsylla) theresae* Durden and Beaucournu, 2006]
Macrostylophora (subgenus not specified) *durdeni* Beaucournu and Wells, 2009
Nycteridopsylla quadrispina Lu and Wu, 2003 is a junior synonym of *Nycteridopsylla iae* Beaucournu and Kock, 1992 (Hastriter, Bush, Dittmar, Hla Bu, and Whiting, 2009)
Paractenopsyllus (Paractenopsyllus) duplantieri Duchemin, 2004
Paractenopsyllus (Paractenopsyllus) genelli Duchemin, 2004
Paractenopsyllus (Consobrinopsyllus) goodmani Duchemin, 2003
Paractenopsyllus (Paractenopsyllus) juliamarinus Duchemin, 2004
Paractenopsyllus (Paractenopsyllus) madagascarensis Hastriter and Dick, 2009
Paractenopsyllus (Paractenopsyllus) oconnori Duchemin and Ratovonjato, 2004
Paractenopsyllus (Paractenopsyllus) raxworhtyi Duchemin and Ratovonjato, 2004
Paractenopsyllus (Paractenopsyllus) ratovonjatoi Duchemin, 2004
Paractenopsyllus (Paractenopsyllus) rouxi Duchemin, 2004
Rhadinopsylla (Actenophthalmus) dinaromydis Brelah and Trilar, 2000
Rhadinopsylla (Actenophthalmus) eotaenomus Liu, et al., 2007
Rhadinopsylla (Actenophthalmus) rotunditruncata Wu, Li and Cai, 1999
Rhinolophopsylla traubi Hastriter, 2009
Stenoponia dabashawensis Zhang and Yu, 1991
Stenoponia shanghaiensis Liu and Wu, 1960
Syngenopsyllus calceatus remotus Li, Hai and Pan, 1991
Tsaractenus rodhaini Duchemin, 2003

* Robert E. Lewis printed his last edition of a world species list in Lewis, R.E. 2009. Siphonaptera, 16th ed. Unpublished list, 1 November 2009.

The current total number of flea species was counted at 2,183 species by Terry Galloway, although he adds that "It took quite a lot of effort to reach the total I did, and I'm still not confident it tells the whole story". Terry noted that "a goal for *Flea News* might be to have subscribers submit names and references for new species as they are described so a running total can be managed more easily and published in each issue so people can see what has been submitted and add to the list where species have been missed."

In order to support this effort, if any subscribers send me new flea descriptions, I will attempt to feature them in *Flea News* for collating into a listing of all flea species on the *Flea News* Network website http://esanetworks.org/.

Please send any corrections to the Editor, R.L. Bossard (bossardtech@gmail.com). I thank everyone who contributed to this discussion.

R.L. Bossard
Editor, *Flea News*

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The 2016 Livestock Insect Workers Conference will be held in Oklahoma City, OK, June 26-29. For further information and deadlines, please check the website http://livestockbugs.okstate.edu/60th-annual-liwc .

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New Dissertation and Thesis

Student: van der Mescht, L.
Title: Exploring mechanisms that shape Siphonaptera composition and distribution patterns on small mammals across South Africa
Degree: Doctor of Science in Conservation Ecology
Date: December 2015
Promoters: Dr. Sonja Matthee, Prof. Conrad A. Matthee
Faculty of AgriSciences, Stellenbosch University

Fleas (Siphonaptera) are obligate ectoparasites of mammals and birds. Due to their economic importance as disease vectors, most contemporary studies on macroparasite now also consider the spatial variation of parasite communities and the underlying mechanisms involved in shaping current distribution patterns. Fleas differ in life history traits, such as the level of host specificity and microhabitat preferences, which can result in differential evolutionary responses to similar abiotic events.

The main objectives of this study are to investigate: (1) the influence of vicariance and host association on the genetic structure of two generalist flea species, *Listropsylla agrippinae* and *Chiastopsylla rossi*. The taxa differ in the time spent on the host (fur vs. nest) and the level of host specificity; (2) the taxonomy of *Dinopsyllus ellobius* to determine whether more than one species exist and if so to then elucidate the mechanisms of speciation; and (3) the influence of flea life history on species distribution model performance and see if the relative importance of predictor variables differ between species with different life histories.

A total of 1423 small mammals were brushed to collect 2906 flea individuals originating from 31 geographically distinct localities throughout South Africa (SA). Phylogeographic structure of *L.*

agrippinae and *C. rossi* were determined by making use of 315 mitochondrial COII (mtDNA) and 174 nuclear EF1-α (nDNA) sequences.

The more host specific fur flea, *L. agrippinae,* displayed pronounced spatial phylogeographic structure, based on mtDNA, which was congruent with host vicariance in the region. In contrast, the more generalist nest flea, *C. rossi*, showed a higher level of inter-populational divergence, based on mtDNA and nDNA, and this may be attributed to comparatively higher restrictions to dispersal when compared to the more specific fur flea.

In an attempt to resolve the taxonomy of *D. ellobius*, 151 mtDNA and 68 nDNA alleles were generated from individuals meeting the morphological description of D. *ellobius.* Two distinct *D. ellobius* lineages that corresponded to previously described species (*D. ellobius* and *D. abaris*) were recorded. The results indicate that the two species indeed differ morphologically and based on the distribution of the two species it was concluded that the diversification could be a result of climate driven vicariance and subsequent ecological segregation according to habitat use. Locality records from Segerman (1995) were digitized and used as background data in species distribution modelling.

Sufficient information was obtained for 21 flea species. A total of 68 climatic and landscape feature predictor variables were obtained and through a process of elimination, 19 variables were ultimately used. Model performance was good to excellent on average and the contribution of climate and landscape feature variables differed between fleas with different life histories.

Historical and contemporary climate has the most prominent effect on flea distribution at the regional scale, but the level of host association influences the phylogeographic pattern of fleas. This study provides the first evidence of congruent phylogeographic patterns between a generalist temporary parasite and its hosts. Our findings provide further support for the notion that more than one species exist within the *D. ellobius* complex and that speciation is a result of complex interactions. The study also provides novel data on the role of environmental variables in shaping the geographic distribution of flea species with different life histories. With the anticipated rise in flea-borne diseases worldwide, due to changes in vector distribution, the study further emphasize the need for studying the mechanisms involved in shaping flea distribution patterns.

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Student: Joseph A. Legan
Title: The Medical Response to the Black Death
Date: Spring 2015
Degree Name: Bachelor of Arts (BA) Thesis
Department: Department of History, James Madison University
First Advisor: John Butt

This paper discusses the medical response to the Black Death in both Europe and the Middle East. The Black Death was caused by a series of bacterial strands [strains] collectively known as *Yersinia pestis*. The Plague originated in the Mongolian Steppes. It was spread westward by the east-west trading system. Once it arrived in the Crimea in 1346, Italian merchants helped spread it throughout the Mediterranean. Medicine in Europe and the Middle East were centered on Galen's theory of humors. There were many religious explanations for the Plague, but the main medical explanation was the spread of bad air, or miasma. Many preventative measures dealt with eliminating the miasma. The three main diagnostic methods used by physicians were astrology, uroscopy, and pulse-taking. Europeans realized the contagious nature of the disease, but many Muslims refuted the notion of contagion. Most cures for the Plague dealt with balancing body humors, such as bloodletting. Other cures included gold, rose water, and theriac. Even though the Plague killed many, it had beneficial effects on medicine, especially in Europe. Doctors began to question Galenic medicine, they relied more on observation, and they paid more attention to anatomy. There were also improvements in medical ethics, public health, and hospitals. Legan, Joseph A., "The Medical Response to the Black Death" (2015). Senior Honors Projects. Paper 103.

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Meetings

There were a couple of ESA presentations on fleas at this year's ESA meeting (Synergy in Science: Partnering for Solutions. 2015 Meeting of the Entomological Society of America, November 15-18, Minneapolis, MN).

Ectoparasite loads and their implications from rats trapped in New Orleans, LA.

Claudia Riegel, Eric Guidry, Anna Peterson, Bruno Ghersi, Friederike Bauder, Timothy Madere, and Michael J. Blum. City of New Orleans, LA and Tulane Univ., New Orleans, LA

**

Species composition and plague prevalence in fleas collected from small mammals in mixed grass prairies: implications for the maintenance of enzootic plague.

Lauren Paul Maestas and Hugh Britten, Univ. of South Dakota, Vermillion, SD

***

New Book

Title: *Parasite Diversity and Diversification: Evolutionary Ecology Meets Phylogenetics*

Authors: S. Morand; Boris R. Krasnov; D.T.J. Littlewood

Publisher: Cambridge University Press, United Kingdom, 2015.

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Featured Research

New Fleas 2015

Atopopsyllus cionus, n. gen., n. sp. Poinar, 2015
Ctenidiosomus austrinus Lopez-Berrizbeitia, Hastriter, and Diaz (in Lopez-Berrizbeitia, Hastriter, Barquez, and Diaz, 2015)
*Plocopsylla linardi*i Sanchez, Beaucournu and Lareschi, 2015

Traubia (Traubia) durdeni Hastriter, 2015
Traubia (Traubia) pilgrim Hastriter, 2015
Traubia (Traubia) traubi Hastriter, 2015
Traubia (Traubia) wilhelmensis Hastriter, 2015
Traubia (Kuruopsylla) n. subgen. Hastriter, 2015
Traubia (Kuruopsylla) eaglini Hastriter, 2015
Traubia (Kuruopsylla) mulcahyi Hastriter, 2015
Traubia (Kuruopsylla) o'harai Hastriter, 2015
Traubia (Kuruopsylla) ungerechti Hastriter, 2015

Beaucournu J.-C., Mutombo P. & Laudisoit A., 2015. Un nouvel *Afristivalius* de la République démocratique du Congo (Siphonaptera, Pygiopsyllidae, Stivaliinae). Bulletin de la Société entomologique de France, sous presse.

Poinar, George, Jr. (2015) A New Genus of Fleas with Associated Microorganisms in Dominican Amber. Journal of Medical Entomology 52(6)1234-1240. 10.1093/jme/tjv134

A flea preserved in Dominican amber is described as *Atopopsyllus cionus, n. gen., n. sp.* (Atopopsyllini n. tribe, Spilopsyllinae, Pulicidae). The male specimen has two unique characters that have not been noted in previous extant or extinct fleas, thus warranting its tribal status. These characters are five-segmented maxillary palps and cerci-like organs on abdominal tergite X. Additional characters are the absence of ctenidia, very small eyes, a lanceolate terminal segment of the maxillary palps, legs with six notches on the dorsal margin of the tibiae, five pairs of lateral plantar bristles on the distitarsomeres, and nearly straight ungues with a wide space between the basal lobe and tarsal claw. Trypanosomes and coccobacilli in the rectum and coccobacilli on the tip of the epipharynx of the fossil are depicted and briefly characterized.

Hosts and Distribution

Acosta, R., Fernández, J.A. (2015) Flea diversity and prevalence on arid-adapted rodents in the Oriental Basin, Mexico. Revista Mexicana de Biodiversidad, 86 (4), pp. 981-988. DOI: 10.1016/j.rmb.2015.09.014

Beaucournu J.-C. & Gomez-Lopez M.S. 2015. Orden Siphonaptera, in Revista Ibero Diversitad Entomologica, SEA, n° 61 A, 1-11.

Beaucournu, Jean-Claude, Prie, Vincent, Ramasindrazana, Beza, Goodman, Steven M., Laudisoit, Anne. 2015. Les Puces de Chauve-souris a Madagascar : nouvelles donnees et clef d'identification illustree actualisee (Siphonaptera, Ischnopsyllidae). [Bat fleas from Madagascar: new data and updated illustrated identification key (Siphonaptera, Ischnopsyllidae)).] Bulletin de la Societe Entomologique de France 120 (3) 323-334.

Beaucournu J.-C., Randrenjarison Andriniaina & Goodman S.M., 2015. Puces d'Ambohitantely, Madagascar: Spécificité et phénologie. Malagasy Nature, 9: 39-48.

Goodman, S.M., Andriniaina, H.R.R., Soarimalala, V., Beaucournu, J.-C. (2015) The Fleas of Endemic and Introduced Small Mammals in Central Highland Forests of Madagascar: Faunistics, Species Diversity, and Absence of Host Specificity. Journal of Medical Entomology 52 (5), pp. 1135-1143. DOI: 10.1093/jme/tjv113

Data are presented on the flea species of the genera *Paractenopsyllus* (Ceratophyllidae, Leptopsyllinae) and *Synopsyllus* (Pulicidae, Xenopsyllinae) obtained from small mammals during two 2014 seasonal surveys at a montane humid forest site (Ambohitantely) in the Central Highlands of Madagascar. The mammal groups included the endemic family Tenrecidae (tenrecs) and subfamily Nesomyinae (rodents) and two introduced families Muridae (rodents) and Soricidae (shrews); no fleas were recovered from the latter family. The surveys were conducted at the end of the wet and dry seasons with 288 individual small mammals captured, including 12 endemic and four introduced species. These animals yielded 344 fleas, representing nine species endemic to Madagascar; no introduced species was collected. Some seasonal variation was found in the number of trapped small mammals, but no marked difference was found in species richness. For flea species represented by sufficient samples, no parasite-host specificity was found, and there is evidence for considerable lateral exchange in the local flea fauna between species of tenrecs and the two rodent families (endemic and introduced). The implications of these results are discussed with regards to small mammal species richness and community structure, as well as a possible mechanism for the maintenance of sylvatic cycles of bubonic plague in the montane forests of Madagascar.

Christensen, N.D., Skirnisson, K., Nielsen, Ó.K. (2015) The parasite fauna of the Gyrfalcon (*Falco rusticolus*) in Iceland. Journal of Wildlife Diseases, 51 (4), pp. 929-933. DOI: 10.7589/2015-01-022

We examined 46 Gyrfalcon (*Falco rusticolus*) carcasses from Iceland for parasites, including 29 first-year birds and 17 second-year birds and older. Endoparasites observed were the trematodes *Cryptocotyle lingua* (prevalence 8%), *Cryptocotyle concavum* (4%), and *Strigea* sp. (8%); the cestode *Mesocestoides* sp. (27%); and the nematodes *Eucoleus contortus* (76%) and *Serratospiculum guttatum* (7%). Ectoparasites included the astigmatan mite *Dubininia accipitrina* (47%), a mesostigmatan rhynonyssid mite (4%), the tick *Ixodes caledonicus* (20%), the mallophagans *Degeeriella rufa* (90%) and *Nosopon lucidum* (7%), the flea *Ceratophyllus vagabundus* (7%), and the louse fly *Ornithomya chloropus* (7%). *Cryptocotyle lingua*, *C. concavum*, *S. guttatum*, *D. accipitrina*, *I. caledonicus*, and *N. lucidum* are new host records. Of the five most common parasites (prevalence >= 20%) only *Mesocestoides* sp. showed a significant age relationship, being more prevalent in adult falcons (P=0.021). *Eucoleus contortus* was also more prevalent in adults with marginal statistical significance (P=0.058). Frounce, caused by *E. contortus* (possibly also by *Trichomonas gallinae*, which was not searched for in the survey) was highly prevalent (43%), but did not show a relationship with host age (P=0.210). Birds with frounce were in poorer body condition than healthy birds (P=0.015).

Mori, Emiliano; Sforzi, Andrea; Menchetti, Mattia; Mazza, Giuseppe; Lovari, Sandro; Pisanu, Benoit. 2015. Ectoparasite load in the crested porcupine *Hystrix cristata* Linnaeus, 1758 in Central Italy. Parasitology Research 114 (6) 2223-2229. 10.1007/s00436-015-4413-3

The crested porcupine *Hystrix cristata* is a large body-sized rodent, occurring in Europe only in the Italian Peninsula, where it may have been introduced in early Medieval times. Its parasite fauna is currently poorly known and limited to few anecdotal observations. We have analyzed the ectoparasite load of 165 crested porcupines from Tuscany and Latium (Central Italy). Both captured and road-killed individuals were checked for fleas and ticks. Overall, only 39 porcupines were infested by four species of ticks and five of fleas. Abundance of ectoparasites was higher in areas with higher habitat richness, with respect to densely wooded areas. The most frequent species was the flea *Pulex irritans* (25%), whose prevalence peaked in winter probably because of optimal abiotic conditions in the porcupine's den. The remaining species of both hard ticks (*Rhipicephalus bursa*, *Pholeoixodes hexagonus*, and *Ixodes ventalloi*) and fleas (*Paraceras melis*, *Ctenocephalides canis*, *Dasypsyllus gallinulae*, and *Hystrichopsylla talpae*), all with prevalence lower than 5 %, could be due to den sharing with other vertebrates, mainly carnivores such as, e.g., red foxes and badgers. The second most prevalent species was the generalist tick *Ixodes ricinus* (21 %). An adult male-biased parasitism for ticks has been detected, suggesting a possible role of testosterone related immune-depressive effect. The low richness in dominant ectoparasite species, built up by locally acquired generalist taxa, provides support to the allochthonous origin of this rodent in Italy.

Lipatova, I.; Stanko, M.; Paulauskas, A.; Spakovaite, S.; Gedminas, V. T. 2015. Fleas (Siphonaptera) in the Nests of Dormice (Gliridae: Rodentia) in Lithuania. Journal of Medical Entomology 52 (3) 469-474. 10.1093/jme/tjv033

Negative effects of flea (Siphonaptera) parasitism on the host may be expressed in different ways. The aim of this study was to assess distribution of the flea fauna in nests of dormice in Lithuania. Nests of *Glis glis* (L.), *Dryomys nitedula* (Pallas), and *Muscardinus avellanarius* (L.) were collected from nest boxes in 2012 and 2013. Fleas were collected from nests in the laboratory and put into plastic tubes with 70% ethanol. Flea species were identified using morphological keys. From 400 nest boxes, 112 nests of dormice were collected from eight sites from mixed forests of central Lithuania. Twenty-three nests of *G. glis* were collected from nest boxes, with 16 of them containing 286 fleas belonging to four species: *Ceratophyllus sciurorum* (Schrank) (259), *C. gallinae* (Schrank) (23), *Hystrichopsylla talpae* (Curtis) (3), and *Megabothris turbidus* (Rothschild) (1). Fourteen nests of *M. avellanarius* were collected from nest boxes, 4 of which contained 224 fleas belonging to two species: *C. sciurorum* (221) and *C. gallinae* (3). Twenty-four nests of *D. nitedula* were collected from nest boxes, including 17 containing 207 fleas belonging to two species: *C. sciurorum* (205) and *C. gallinae* (2). Fifty-one nests of undetermined dormice species also were collected from nest boxes, 12 of them contained 395 fleas belonging to three species: *C. sciurorum* (374), *Ctenophthalmus agyrtes* (Heller) (19), and *Ctenophthalmus assimilis* (Taschenberg) (2). C. sciurorum was a predominant species in the nests of dormice. The occurrence of *C. gallinae* was documented in Lithuania for the first time.

Morphology and Physiology

Medvedev, S.G. (2015) Morphological diversity of the skeletal structures of fleas (Siphonaptera). Part 1: the general characteristic and features of the head. (2015) Entomological Review, 95 (7), pp. 852-873. DOI: 10.1134/S0013873815070040

Analysis of a considerable amount of data on the anatomy of 14 structures of the head capsule in 96 genera of fleas (over 90% of the genera in the world fauna) shows that different flea taxa can be described based on 32 *universal* and *specific* characters, whose 135 states reflect the entire known diversity of the flea head morphology. Of them, 17 characters can be formulated based on *universal* terms applicable to any structure. Elementary characters can reflect phylogenetic affinity only in association with *specific* characters; these 15 characters are formulated based on specific features of the flea morphology and describe interrelations of the skeletal structures (or microstructures of a particular formation). The concept of a "structural type" is used for multi-aspect characteristics of different character states of some structures. Homoplasies at various levels comprise more than half (53%) of all the character states in the head structure. The character states reflecting the phylogenetic closeness of taxa make up about 22%. DOI: 10.1134/S001387381506001X

Downs, Cynthia J., Pinshow, Berry, Khokhlova, Irina S., Krasnov, Boris R. (2005) Flea fitness is reduced by high fractional concentrations of CO_2 that simulate levels found in their hosts' burrows. Journal of Experimental Biology 218(22)3596-3603.

Nidicolous ectoparasites such as fleas and gamasid mites that feed on small and medium-sized mammals spend much of their time in their hosts' burrows, which provide an environment for living, and often feeding, to their pre-imaginal and/or adult stages. Thus, these ectoparasites should be adapted to environmental conditions in burrows, including high fractional concentrations of CO_2 (F-CO_2). We examined how a high F-CO_2 (0.04) affected survival and reproductive success of a hematophagous ectoparasite of burrowing rodents using fleas *Xenopsylla ramesis* and Sundevall's jirds *Meriones crassus*. In the first experiment, fleas fed on hosts housed in high-CO_2 (F-CO_2 = 0.04) or atmospheric-CO^2 (F-CO_2 approximate to 0.0004) air, and were allowed to breed. In a second experiment, fleas were maintained in high CO_2 or CO_2 -free air with no hosts to determine how CO_2 levels affect survival and activity levels. We found that at high F-CO_2 fleas laid fewer eggs, reducing reproductive success. In addition, at high F-CO_2, activity levels and survival of fleas were reduced. Our results indicate that fleas do not perform well in the F-CO_2 used in this experiment. Previous research indicated that the type and intensity of the effects of CO_2 concentration on the fitness of an insect depend on the F-CO_2 used, so we advise caution when generalizing inferences drawn to insects exposed to other F-CO_2. If, however, F-CO_2 found in natural mammal burrows brings about reduced fitness in fleas in general, then burrowing hosts may benefit from reduced parasite infestation if burrow air F-CO_2 is high.

Kernif, T; Stafford, K; Coles, G C; Bitam, I; Papa, K; Chiaroni, J; Raoult, D; Parola, P. 2015. Responses of artificially reared cat fleas *Ctenocephalides felis felis* (Bouche, 1835) to different mammalian bloods. Medical and veterinary entomology 29 (2) 171-7. 10.1111/mve.12100

The cat flea, *Ctenocephalides felis felis* (Bouche, 1835) (Siphonaptera: Pulicidae), which is found worldwide and which parasitizes many species of wild and domestic animal, is a vector and/or reservoir of bacteria, protozoa and helminths. To aid in the study of the physiology and behaviour of fleas and of their transmission of pathogens, it would be of value to improve the laboratory rearing of pathogen-free fleas. The conditions under which artificially reared fleas at the University of Bristol (U.K.) and the Rickettsial Diseases Institute (France) are maintained were studied, with different ratios of male to female fleas per chamber (25 : 50, 50 : 100, 100 : 100, 200 : 200). The fleas were fed with bovine, ovine, caprine, porcine or human blood containing the anticoagulants sodium citrate or EDTA. Egg production was highest when fleas were kept in chambers with a ratio of 25 males to 100 females.

In addition, the use of EDTA as an anticoagulant rather than sodium citrate resulted in a large increase in the number of eggs produced per female; however, the low percentage of eggs developing through to adult fleas was lower with EDTA. The modifications described in our rearing methods will improve the rearing of cat fleas for research.

[Editor's comment: Artificial hosts were discussed in previous *Flea News* (71 and 72) as a technique to compare the factors that make a host favorable to a flea. Though this research did not include the blood of cat or dog (the usual hosts of *C. felis*), the research shows that different bloods have significant effects on flea development and egg production, presumably due to unknown factors in blood.]

Ecology and Evolution

Sponchiado, J., Melo, G.L., Landulfo, G.A., Jacinavicius, F.C., Barros-Battesti, D.M., Cáceres, N.C. (2015) Interaction of ectoparasites (Mesostigmata, Phthiraptera and Siphonaptera) with small mammals in Cerrado fragments, western Brazil. (2015) Experimental and Applied Acarology, 66 (3), pp. 369-381. DOI: 10.1007/s10493-015-9917-0

Paz, G.F., Avelar, D.M., Reis, I.A., Linardi, P.M. (2015) Dynamics of *Ctenocephalides felis felis* (Siphonaptera: Pulicidae) Infestations on Urban Dogs in Southeastern Brazil. (2015) Journal of Medical Entomology, 52 (5), pp. 1159-1164. DOI: 10.1093/jme/tjv071

The cat flea, *Ctenocephalides felis felis* (Bouché, 1835), is an important ectoparasite of dogs and cats throughout the world, causing annoyance to the animals and acting as a vector of infections and a cause of allergic dermatitis in dogs and cats. Although climatic variability and seasonality are known to influence the diversity and abundance of fleas, few investigations of seasonal prevalence of cat flea infestation have involved the same group of dogs being examined regularly over an extended period. The aim of this study was to determine the effects of temperature, rainfall, and relative humidity on the infestation by *C. felis felis* on 88 outdoor dogs in southeastern Brazil. The dogs, which were of mixed breed, sex, and age, were examined for ectoparasites every month during the period August 2011 to July 2012, and samples of fleas were randomly collected and identified. Meteorological data, comprising mean temperature, total rainfall, and mean relative humidity, were recorded for the calendar month prior to that in which the examinations were performed. Dogs were found to be infested only with *C. felis felis*, with a higher prevalence in the months with lowest rainfall (July, August, and September). The data obtained in this investigation can be used in control programs in order to establish an efficient strategy for environmental management and the application of insecticides, particularly during the driest months of the year based on the seasonal pattern of infestation of dogs by *C. felis felis*.

Levick, Bethany Laudisoit, Anne Wilschut, Liesbeth Addink, Elisabeth Ageyev, Vladimir Yeszhanov, Aidyn Sapozhnikov, Valerij Belayev, Alexander Davydova, Tania Eagle, Sally Begon, Mike. 2015. The Perfect Burrow, but for What? Identifying Local Habitat Conditions Promoting the Presence of the Host and Vector Species in the Kazakh Plague System. PLoS One 10 (9) e0136962 – e0136962. DI 10.1371/journal.pone.0136962

The wildlife plague system in the Pre-Balkhash desert of Kazakhstan has been a subject of study for many years. Much progress has been made in generating a method of predicting outbreaks of the disease (infection by the gram negative bacterium *Yersinia pestis*) but existing methods are not yet accurate enough to inform public health planning. The present study aimed to identify characteristics of individual mammalian host (*Rhombomys opimus*) burrows related to and potentially predictive of the presence of *R. opimus* and the dominant flea vectors (*Xenopsylla* spp.). Methods Over four seasons, burrow characteristics, their current occupancy status, and flea and tick burden of the occupants were recorded in the field. A second data set was generated of long term occupancy trends by recording the occupancy status of specific burrows over multiple occasions. Generalised linear mixed models were constructed to identify potential burrow properties predictive of either occupancy or flea burden. Results At the burrow level, it was identified that a burrow being occupied by *Rhombomys*, and remaining occupied, were both related to the characteristics of the sediment in which the burrow was constructed. The flea burden of *Rhombomys* in a burrow was found to be related to the tick burden. Further larger scale properties were also identified as being related to both *Rhombomys* and flea presence, including latitudinal position and the season. Conclusions Therefore, in advancing our current predictions of plague in Kazakhstan, we must consider the landscape at this local level to increase our accuracy in predicting the dynamics of gerbil and flea populations. Furthermore this demonstrates that in other zoonotic systems, it may be useful to consider the distribution and location of suitable habitat for both host and vector species at this fine scale to accurately predict future epizootics.

Raveh, S., Neuhaus, P., Dobson, F.S. (2015) Ectoparasites and fitness of female Columbian ground squirrels. (2015) Philosophical Transactions of the Royal Society B: Biological Sciences, 370 (1669), 9 p. DOI: 10.1098/rstb.2014.0113

Parasites play an important role in the evolution of host traits via natural selection, coevolution and sexually selected ornaments used in mate choice. These evolutionary scenarios assume fitness costs for hosts. To test this assumption, we conducted an ectoparasite removal experiment in free-living Columbian ground squirrels (*Urocittelus columbianus*) in four populations over three years. Adult females were randomly chosen to be either experimentally treated with anti-parasite treatments (spot-on solution and flea powder, N = 61) or a sham treatment (control, N = 44). We expected that experimental females would show better body condition, increased reproductive success and enhanced survival. Contrary to our expectations, body mass was not significantly different between treatments at mating, birth of litter or weaning of young. Further, neither number nor size of young at weaning differed significantly between the two treatments. Survival to the next spring for adult females and juveniles was not significantly different between experimental and control treatments. Finally, annual fitness was not affected by the treatments. We concluded that females and their offspring were able compensate for the presence of ectoparasites, suggesting little or no fitness costs of parasites for females in the different colonies and during the years of our experiments.

Kumar, Arvind. 2015. SEASONAL ABUNDANCE OF *Ctenocephalides canis* (CURTIS). Journal of Experimental Zoology India 18 (1)245-247.

Surveillance on *Ctenocephalides canis* was conducted from January 2012 to December 2012 in

Baraut town of Uttar Pradesh, India. A total of 67 dogs were searched for fleas. The higher positivity of dogs was recorded in the month of March as 100% while lowest in October and November. The highest flea index of 5.4 was recorded in May and lowest in December and January. The male female ratio was 1:5.00 in April and 1:1 in December and January. The analysis of data shows increasing flea prevalence at the onset of summer and minimum in winter.

Control

Rust, M.K., Vetter, R., Denholm, I., Blagburn, B., Williamson, M.S., Kopp, S., Coleman, G., Hostetler, J., Davis, W., Mencke, N., Rees, R., Foit, S., Böhm, C., Tetzner, K. (2015) Susceptibility of Adult Cat Fleas (Siphonaptera: Pulicidae) to Insecticides and Status of Insecticide Resistance Mutations at the Rdl and Knockdown Resistance Loci. Parasitology Research, 114, pp. 7-18. DOI: 10.1007/s00436-015-4512-1

The susceptibility of 12 field-collected isolates and 4 laboratory strains of cat fleas, *Ctenocephalides felis* was determined by topical application of some of the insecticides used as on-animal therapies to control them. In the tested field-collected flea isolates the LD50 values for fipronil and imidacloprid ranged from 0.09 to 0.35 ng/flea and 0.02 to 0.19 ng/flea, respectively, and were consistent with baseline figures published previously. The extent of variation in response to four pyrethroid insecticides differed between compounds with the LD50 values for deltamethrin ranging from 2.3 to 28.2 ng/flea, etofenprox ranging from 26.7 to 86.7 ng/flea, permethrin ranging from 17.5 to 85.6 ng/flea, and d-phenothrin ranging from 14.5 to 130 ng/flea. A comparison with earlier data for permethrin and deltamethrin implied a level of pyrethroid resistance in all isolates and strains. LD50 values for tetrachlorvinphos ranged from 20.0 to 420.0 ng/flea. The rdl mutation (conferring target-site resistance to cyclodiene insecticides) was present in most field-collected and laboratory strains, but had no discernible effect on responses to fipronil, which acts on the same receptor protein as cyclodienes. The kdr and skdr mutations conferring target-site resistance to pyrethroids but segregated in opposition to one another, precluding the formation of genotypes homozygous for both mutations.

Microbial Symbiotes

Fernández-González, A. M., Kosoy, M. Y., Rubio, A. V., Graham, C. B., Montenieri, J. A., Osikowicz, L. M., Y. Bai, R. Acosta-Gutierrez, ... & Suzán, G. (2015). Molecular Survey of *Bartonella* Species and *Yersinia pestis* in Rodent Fleas (Siphonaptera) From Chihuahua, Mexico. Journal of medical entomology. doi: 10.1093/jme/tjv181

Suntsov, V.V. (2015) On the origin of *Yersinia pestis*, a causative agent of the plague: A concept of population-genetic macroevolution in transitive environment. Zhurnal obshcheĭ biologii, 76 (4), pp. 310-318.

An ecological scenario is proposed for the origin of causative agent of the plague (the bacterium *Yersinia pestis*) from the clone of pseudotuberculous microbe of the first serotype *Y. pseudotuberculosis* O:1b. Disclosed are the conditions of gradual intrusion of psychrophile saprozoonosis ancestor into the blood of the primary host, Mongolian tarbagan marmot *Marmota sibirica*. As an inductor of speciation acted the Sartan cooling that occurred in the end of late

Pleistocene under conditions of arid ultra-continental climate in Central Asia. Soil freezing down to the level of hibernating chambers in marmot burrows initiated the transition of marmot flea, *Oropsylla silantiewi*, larvae to optional hemophagy on the mucous coat inside the mouth cavity of sleeping marmots. In its turn, this promoted the conditions of mass traumatic intrusion of Y pseudotuberculosis into marmots bloodstream from faecal particles getting in their mouth cavity in course of building up a plug in a burrow for hibernating. In marmot populations, the selection of bacteria underwent under conditions of heterothermy with repeated changes of hibernating marmots body temperature within the range of 5-37 degrees C (torpor-euthermy). During the warm season, when pseudotuberculous microbes are totally eliminated from the bloodstream of healthy marmots with body temperature about 37 degrees C, bacteria could survive in fleas' digestive tract in the form of biofilm developing in proventriculus as a so called blockage. Final isolation between ancestral and daughter species was helped by the development of intrapopulation antagonism related with the beginning of full-scale synthesis of bacteriocin pesticin. Population-genetic processes in the "marmot-flea" system have led to a macroevolutionary event, that is, to passage of bacteria in a new ecological niche and adaptive zone that are principally different from those of the ancestor. All the present intraspecies forms of *Y. pestis* that appeared due to microevolution, have originated with the subspecies *Y. pestis tarbagani* that has formed in Central Asia during the Sartan cooling.

Cohen, C., Toh, E., Munro, D., Dong, Q., Hawlena, H. (2015) Similarities and seasonal variations in bacterial communities from the blood of rodents and from their flea vectors. ISME Journal, 9 (7), pp. 1662-1676. DOI: 10.1038/ismej.2014.255

Vector-borne microbes are subject to the ecological constraints of two distinct microenvironments: that in the arthropod vector and that in the blood of its vertebrate host. Because the structure of bacterial communities in these two microenvironments may substantially affect the abundance of vector-borne microbes, it is important to understand the relationship between bacterial communities in both microenvironments and the determinants that shape them. We used pyrosequencing analyses to compare the structure of bacterial communities in *Synosternus cleopatrae* fleas and in the blood of their *Gerbillus andersoni* hosts. We also monitored the interindividual and seasonal variability in these bacterial communities by sampling the same individual wild rodents during the spring and again during the summer. We show that the bacterial communities in each sample type (blood, female flea or male flea) had a similar phylotype composition among host individuals, but exhibited seasonal variability that was not directly associated with host characteristics. The structure of bacterial communities in male fleas and in the blood of their rodent hosts was remarkably similar and was dominated by flea-borne *Bartonella* and *Mycoplasma* phylotypes. A lower abundance of flea-borne bacteria and the presence of *Wolbachia* phylotypes distinguished bacterial communities in female fleas from those in male fleas and in rodent blood. These results suggest that the overall abundance of a certain vector-borne microbe is more likely to be determined by the abundance of endosymbiotic bacteria in the vector, abundance of other vector-borne microbes co-occurring in the vector and in the host blood and by seasonal changes, than by host characteristics.

Harrison, G.F., Scheirer, J.L., Melanson, V.R. (2015) Development and validation of an arthropod maceration protocol for zoonotic pathogen detection in mosquitoes and fleas. Journal of Vector Ecology, 40 (1), pp. 83-89. : 10.1111/jvec.12136 .

Ying Bai, Maria Fernanda Rizzo, Danilo Alvarez, David Moran, Leonard F. Peruski, and Michael Kosoy. 2015. Coexistence of *Bartonella henselae* and *B. clarridgeiae* in populations of cats and their fleas in Guatemala. Journal of Vector Ecology 40 (2): 327-332.

Cats and their fleas collected in Guatemala were investigated for the presence of *Bartonella* infections. *Bartonella* bacteria were cultured from 8.2% (13/159) of cats, and all cultures were identified as *B. henselae*. Molecular analysis allowed detection of *Bartonella* DNA in 33.8% (48/142) of cats and in 22.4% (34/152) of cat fleas using gltA, nuoG, and 16S–23S internal transcribed spacer targets. Two *Bartonella* species, *B. henselae* and *B. clarridgeiae*, were identified in cats and cat fleas by molecular analysis, with *B. henselae* being more common than *B. clarridgeiae* in the cats (68.1%; 32/47 vs 31.9%; 15/47). The nuoG was found to be less sensitive for detecting *B. clarridgeiae* compared with other molecular targets and could detect only two of the 15 *B. clarridgeiae*-infected cats. No significant differences were observed for prevalence between male and female cats and between different age groups. No evident association was observed between the presence of *Bartonella* species in cats and in their fleas.

María Jesús Gracia, José Miguel Marcén, Rocio Pinal, Carlos Calvete, and Daniel Rodes. 2015. Prevalence of *Rickettsia* and *Bartonella* species in Spanish cats and their fleas. Journal of Vector Ecology 40 (2): 233-239.

The aim of this study was to determine the prevalence of *Bartonella henselae*, *Rickettsia felis*, and *Rickettsia typhi* in fleas and companion cats (serum and claws) and to assess their presence as a function of host, host habitat, and level of parasitism. Eighty-nine serum and claw samples and 90 flea pools were collected. Cat sera were assayed by IFA for *Bartonella henselae* and *Rickettsia* species IgG antibodies. Conventional PCRs were performed on DNA extracted from nails and fleas collected from cats. A large portion (55.8%) of the feline population sampled was exposed to at least one of the three tested vector-borne pathogens. Seroreactivity to *B. henselae* was found in 50% of the feline studied population, and to *R. felis* in 16.3%. *R. typhi* antibodies were not found in any cat. No *Bartonella* sp. DNA was amplified from the claws. Flea samples from 41 cats (46%) showed molecular evidence for at least one pathogen; our study demonstrated a prevalence rate of 43.3 % of *Rickettsia* sp. and 4.4% of *Bartonella* sp. in the studied flea population. None of the risk factors studied (cat's features, host habitat, and level of parasitation) was associated with either the serology or the PCR results for *Bartonella* sp. and *Rickettsia* sp.. Flea-associated infectious agents are common in cats and fleas and support the recommendation that stringent flea control should be maintained on cats.

Peniche-Lara, G., Jimenez-Delgadillo, B., Dzul-Rosado, K. (2015) *Rickettsia rickettsii* and *Rickettsia felis* infection in *Rhipicephalus sanguineus* ticks and *Ctenocephalides felis* fleas co-existing in a small city in Yucatan, Mexico. Journal of Vector Ecology, 40 (2), pp. 422-424. DOI: 10.1111/jvec.12185

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Fleas in Art and History

Shakespeare on Fleas (Part 2)

The tench (*Tinca tinca*) (Cyprinidae) mentioned in *King Henry IV* (see Flea News 76 June 2015) is a Palearctic fish widely distributed in Europe and Asia, and including the River Thames. The tench is tolerant of brackish, deoxygenated water.

Shakespeare's statement of 'being spotted like a tench' in reference to fleabites is obscure. One idea is that the line is simply humorous nonsense. Another idea is that “a tench” is London slang for “a bad person”. A third idea is that tench are covered in red spots, though pictures of tench do not seem to show this condition. A fourth idea is that tench may be bitten by other fish, resulting in loss of scales and giving a spotted condition. Another idea suggests that tench are considered to be 'doctor fish' capable of curing other fishes' wounds.

An additional quote from Shakespeare mentioning fleas:

King Henry V. Act III. Scene 7.

DUKE OF ORLEANS: Foolish curs, that run winking into the mouth of a
Russian bear and have their heads crushed like
rotten apples! You may as well say, that's a
valiant flea that dare eat his breakfast on the lip of a lion.

**

(on the next page)

Mme. de Bouvillon Tempts Fate by Asking Ragotin to Search for a Flea (c. 1730), by Jean-Baptiste Pater (1695-1736) (Valenciennes, France), oil on canvas, 28x38 cm.

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